Report

I have a dashboard for helping banks determine the best people to contact to sell long term bank loans, I have 4 graphs on my dashboard, one contains employment variation rate by consumer confidence index by job type, this graph is displayed as a scatter graph and it has a table above it to show which colours are which job role, this graph also has a trend line. I picked this graph because it was good to see which job types related to the employment variation rate and the consumer confidence index. I also have a graph that shows age by consumer price index, this graph is displayed as a line chart, I chose this chart as I thought it looked the most appropriate to display the results as the line starts of with a peak and then slowly starts to dip. I chose to display this chart because by seeing age by consumer price index you can tell early 30's are the highest consumer price index. My third graph showed month by employment variation rate, this graph is displayed as a line chart as I believed this was the most appropriate way to display this information as the chart dips slightly, rises and then dips again. It is interesting to see which month has the highest employment variation rate to help banks determine that July is the highest employment variation month. My fourth graph I have month by euroibor 3 month rate. This is displayed as a histogram as I have attached a filter function to this graph, aswell as the month by employment variation rate graph, this filter allows the bank to pick which months they want to see to be able to concentrate on that month alone and it will show the information for that month only and you can also select multiple months so the bank can compare two or three months at the same time. From the histogram it is clear may has the highest Euroibor 3 month rate. I have made this graph the biggest as I think this is an important graph, because you can filter and pick the months you want to see it is an important graph to stand out. Euroibor 3 month rate is highest in may meaning that months there probably wouldn't be as much success of selling long term bank loans as people don't want high rates of interest. So the best months to contact people based on euroibor 3 month rate is December. I have the month by employment variation rate graph below this one as they are both used by the same filter so they can be easily compared. I have the other two graphs at the side, visible but not too big as they are not as important. If you right click on a graph a drill down box will appear and this takes you directly to a chart describing how age dependent on job and education level has an effect on how long a call lasts. Job types are dependent on education so the lower education a person has the Lower level of a job they are

going to have, this can effect weather or not a person will buy a long term loan as the ones with a higher education and job type will be able to afford higher repayments than those who are Lower down the chain on education or job type. I also have some text in the middle to not draw to much attention to this as I want people to look at and read the graphs.

I have a filter function as seen in Fig.1. this can be used on the month by euroibor3M histogram as shown below and can also be used on the month by emp.var.rate graph by clicking on the search by month as highlighted below, you click on the arrow to the right side of (All) and it will drop down with all the months, you click the month or months you want to filter the results to and it will show all the data on those two graphs of the only the months you have clicked making it easier to see the data you need to see. I chose to add the filter function to these two graphs as it would be beneficial to

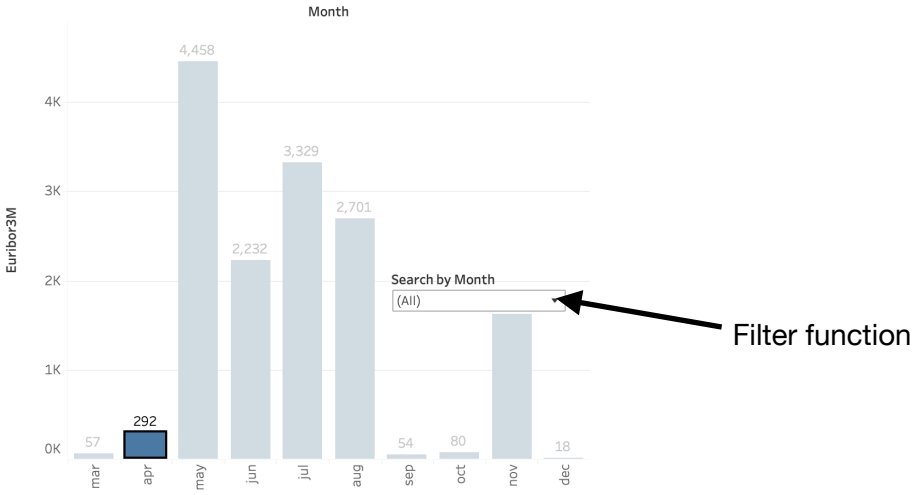
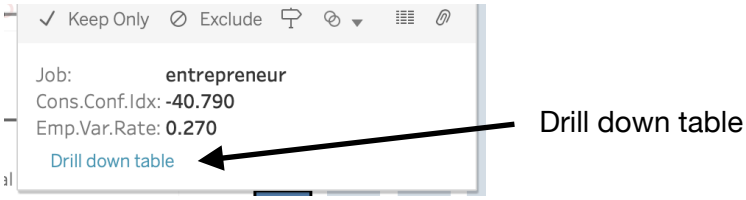


Fig.1.

I also have a drill down function as shown below in Fig.2. This can be used by double clicking or right clicking on any of the graphs data and a drill down table will show at the bottom of the pop up as highlighted below in Fig.2. you then click this drill down table text and it will take you directly to a data table showing call duration based on age, job and education as shown below in Fig.3. These filters make it easy to see extra data without adding to much to the dashboard to make it overcrowded.

Fig.2.



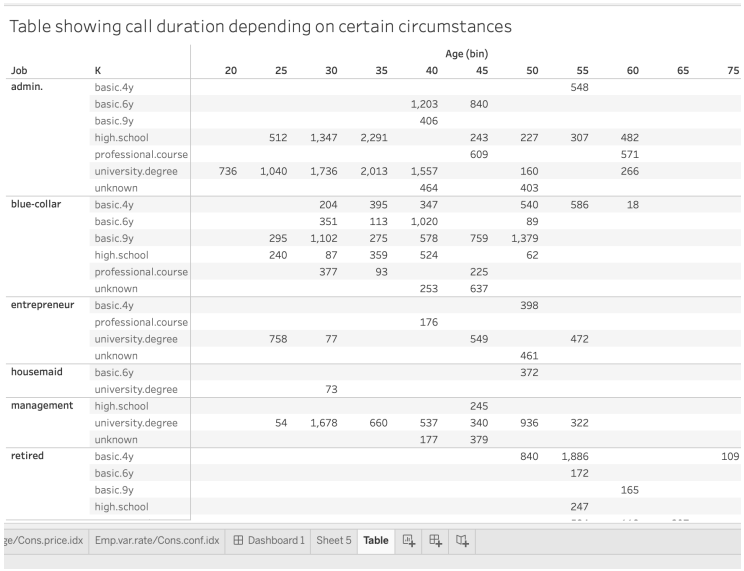


Table showing call duration depending on certain circumstances

Job	K	Age (bin) 20	25	30	35	40	45	50	55	60	65	75
admin.	basic.4y								548			
	basic.6y					1,203	840					
	basic.9y					406						
	high.school		512	1,347	2,291		243	227	307	482		
	professional.course						609			571		
	university.degree	736	1,040	1,736	2,013	1,557		160		266		
	unknown					464		403				
blue-collar	basic.4y			204	395	347		540	586	18		
	basic.6y			351	113	1,020		89				
	basic.9y		295	1,102	275	578	759	1,379				
	high.school		240	87	359	524		62				
	professional.course			377	93		225					
	unknown					253	637					
entrepreneur	basic.4y							398				
	professional.course					176						
	university.degree		758	77			549		472			
	unknown							461				
housemaid	basic.6y							372				
	university.degree			73								
management	high.school						245					
	university.degree		54	1,678	660	537	340	936	322			
	unknown					177	379					
retired	basic.4y							840	1,886			109
	basic.6y								172			
	basic.9y									165		
	high.school								247			

ge/Cons.price.idx | Emp.var.rate/Cons.conf.idx | Dashboard 1 | Sheet 5 | Table

Fig.3.

Overall as dashboard will help banks to look at the important data and be able to pick out the important bits that they can use to determine who to call for a successful call selling long term bank loans. From the data we can see December is the best month to call as it has the lowest interest rate, but July has the highest employment variation rate, December is quite low. Early 30's are the best age group to call based on consumer price index.

References:

Global rates. Euribor 3 months. Available at: https://www.global-rates.com/en/interest-rates/euribor/2/euribor-interest-3-months/#:~:text=The 3-month Euribor interest rate is the interest rate,a maturity of 3 months. [date accessed 31st May 2024].

J. Fernando. (2024) consumer price index (CPI): what it Is and how its used. Available at: https://www.investopedia.com/terms/c/consumerpriceindex.asp [date accessed 30th May 2024].

Tableau. (2024) create a dashboard. Available at: https://help.tableau.com/current/pro/desktop/en-us/dashboards_create.htm [date accessed 01st June 2024].